

JOBSHEET 15

Collection

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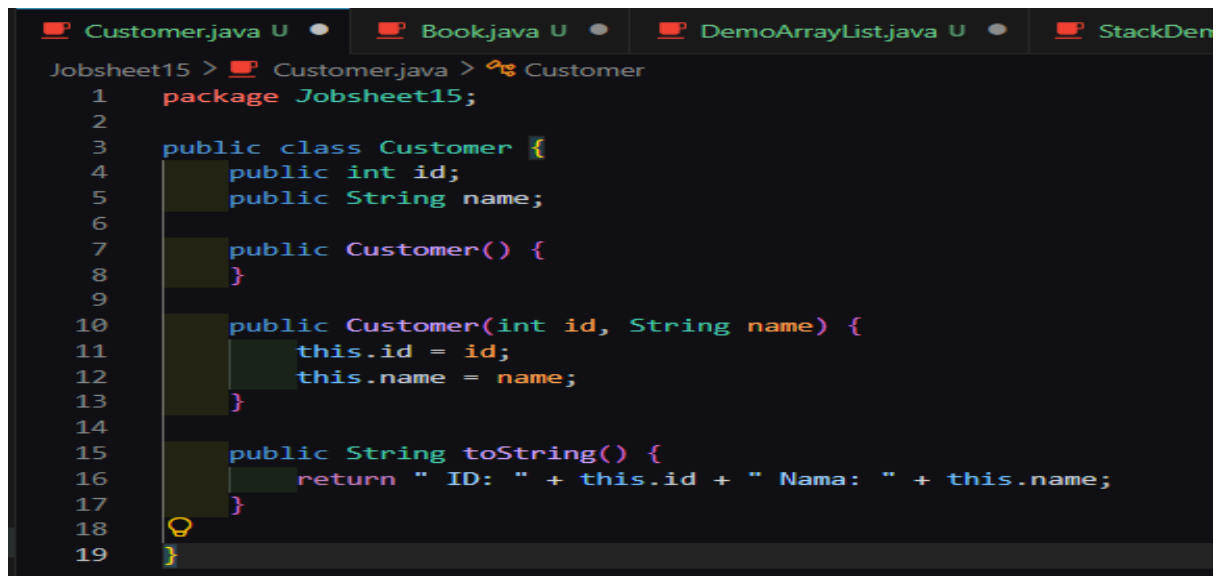
NIM : 254107020019

KELAS : TI 1-G

NO. ABSEN : 18

13.2) Persiapan.

Code Customer.java:



```
Customer.java U • Book.java U • DemoArrayList.java U • StackDen
Jobsheet15 > Customer.java > Customer
1 package Jobsheet15;
2
3 public class Customer {
4     public int id;
5     public String name;
6
7     public Customer() {
8     }
9
10    public Customer(int id, String name) {
11        this.id = id;
12        this.name = name;
13    }
14
15    public String toString() {
16        return " ID: " + this.id + " Nama: " + this.name;
17    }
18
19 }
```

Code book.java:



```
Customer.java U • Book.java U • TreeSetDemo.java U • StackDemo.java U • DemoArrayList.jav
Jobsheet15 > Book.java > ...
1 package Jobsheet15;
2
3 public class Book {
4     public String isbn;
5     public String title;
6
7     public Book() {
8     }
9
10
11    public Book(String isbn, String title) {
12        this.isbn = isbn;
13        this.title = title;
14    }
15
16    public String toString() {
17        return "ISBN: " + this.isbn + " Title: " + this.title;
18    }
19
20 }
21
```

13.3) Praktikum (Implementasi ArrayList).

Code DemoArrayList.java:

```
Customer.java U Book.java U DemoArrayList.java U X
JobSheet15 > DemoArrayList.java > DemoArrayList
1 package Jobsheet15;
2
3 import java.util.ArrayList;
4
5 public class DemoArrayList {
6     public static void main(String[] args) {
7         ArrayList<Customer> customers = new ArrayList<>(initialCapacity: 0);
8         Customer customer1 = new Customer(id: 1, name: "Zakia");
9         Customer customer2 = new Customer(id: 5, name: "Budi");
10
11         customers.add(customer1);
12         customers.add(customer2);
13
14         customers.add(new Customer(id: 4, name: "Cicak"));
15
16         customers.add(index: 2, new Customer(id: 100, name: "Rosa"));
17
18         System.out.println(customers.indexOf(customer2));
19         System.out.println();
20
21         Customer customer = customers.get(index: 1);
22         System.out.println(customer.name);
23         customer.name = "Budi Oetomo";
24
25         ArrayList<Customer> newCustomers = new ArrayList<>();
26         newCustomers.add(new Customer(id: 201, name: "Della"));
27         newCustomers.add(new Customer(id: 202, name: "Victor"));
28         newCustomers.add(new Customer(id: 203, name: "Sarah"));
29
30         customers.addAll(newCustomers);
31
32         System.out.println(customers);
33
34         for (Customer cust : customers) {
35             System.out.println(cust.toString());
36         }
37     }
38 }
```

Outputnya:

```
PS C:\TIA\ASD\PraktikumAlgoritmaStrukturData> & "C:\Program Files\Java\jdk-24\bin\java.exe" "-Xk:+ShowCodeDetailsInExceptionMessages" "-cp" "C:\Users\user\AppData\Roaming\Code\User\workspaceStrukturData_b46f464f\bin" "Jobsheet15.DemoArrayList"
1
Budi
[ ID: 1 Nama: Zakia, ID: 5 Nama: Budi Oetomo, ID: 100 Nama: Rosa, ID: 4 Nama: Cicak, ID: 201 Nama: Della, ID: 202 Nama: Victor, ID: 203 Nama: Sarah]
ID: 1 Nama: Zakia
ID: 5 Nama: Budi Oetomo
ID: 100 Nama: Rosa
ID: 4 Nama: Cicak
ID: 201 Nama: Della
ID: 202 Nama: Victor
ID: 203 Nama: Sarah
PS C:\TIA\ASD\PraktikumAlgoritmaStrukturData>
```

Pertanyaan!

Langkah No.5

Compile dan run kode program, di mana object yang baru ditambahkan? Di awal, di tengah, atau di akhir collection?

Jawaban: Penambahan data akan di tempatkan di akhir collection

Langkah No.7

Compile dan run kode program. Index pada ArrayList dimulai dari 0 atau 1?

Jawaban: Index pada array list dimulai dari 0, dibuktikan dari ketika di print `customers.indexOf(customer2)` yang keluar adalah 1

Langkah No.10

Cobalah hapus angka 2 saat instansiasi object customers. Apakah ArrayList dapat diinstansiasi tanpa harus menentukan size di awal?

Ya, ArrayList bisa di ininstansiasi tanpa adanya size awal

13.4) Praktikum (Implementasi Stack).

Code StackDemo.java:

```
Customer.java U Book.java U DemoArrayList.java U StackDemo.java U
Jobsheet15 > StackDemo.java > StackDemo
1 package Jobsheet15;
2
3 import java.util.Stack;
4
5 public class StackDemo {
6     public static void main(String[] args) {
7         Book book1 = new Book(isbn: "1234", title: "Dasar Pemrograman");
8         Book book2 = new Book(isbn: "7145", title: "Hafalan Sholat Delisa");
9         Book book3 = new Book(isbn: "3562", title: "Muhammad Al-Fatih");
10
11         Stack<Book> books = new Stack<>();
12         books.push(book1);
13         books.push(book2);
14         books.push(book3);
15
16         Book temp = books.peek();
17         if (temp != null) {
18             System.out.println(temp.toString());
19         }
20         Book temp2 = books.pop();
21
22         if (temp2 != null) {
23             System.out.println(temp2.toString());
24         }
25         for (Book book : books) {
26             System.out.println(book.toString());
27         }
28
29         int posisi = books.search(o: "Dasar Pemrograman");
30         System.out.println("Posisi " + posisi);
31     }
32 }
```

Outputnya:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCo
ws\PraktikumAlgoritmaStrukturData_b46f464f\bin' 'Jobsheet15.StackDemo'
ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 1234 Title: Dasar Pemrograman
ISBN: 7145 Title: Hafalan Sholat Delisa
Posisi -1
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>
```

Pertanyaan!

Langkah No.5

Mengapa perlu ada pengecekan (temp != null)?

Jawaban: Untuk melakukan pengecekan bahwa objek temp tidak kosong untuk melakukan fungsi toString.

Langkah No.8

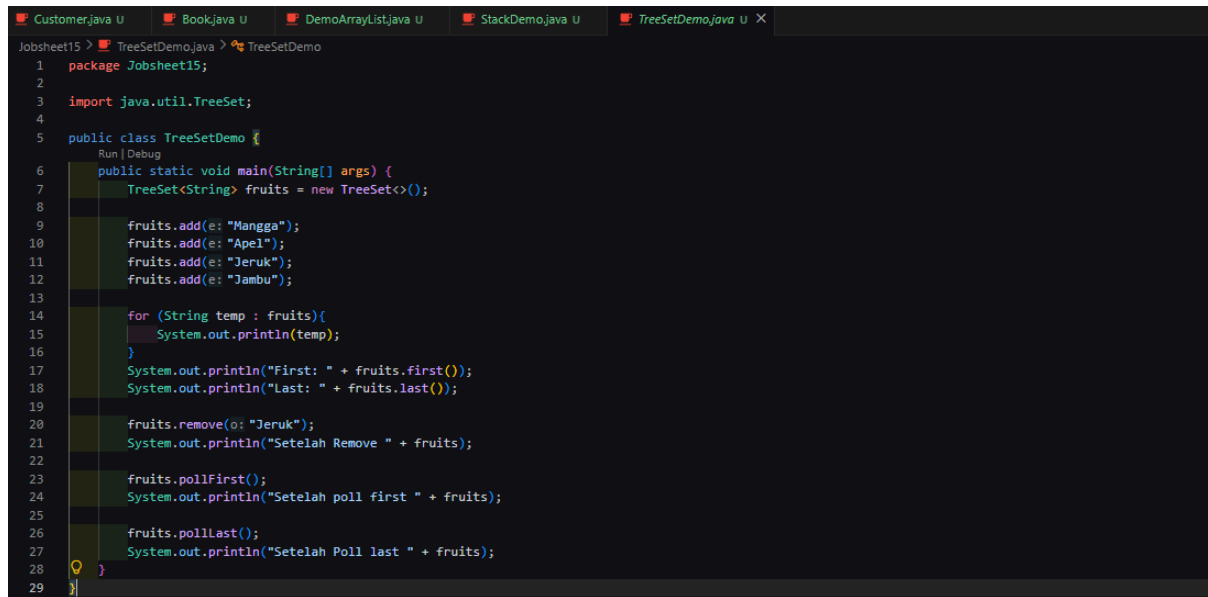
Bagaimana cara melakukan pencarian elemen pada stack menggunakan method search()?

Jawaban: int posisi = books.search("Dasar Pemrograman");

System.out.println("Posisi " + posisi);

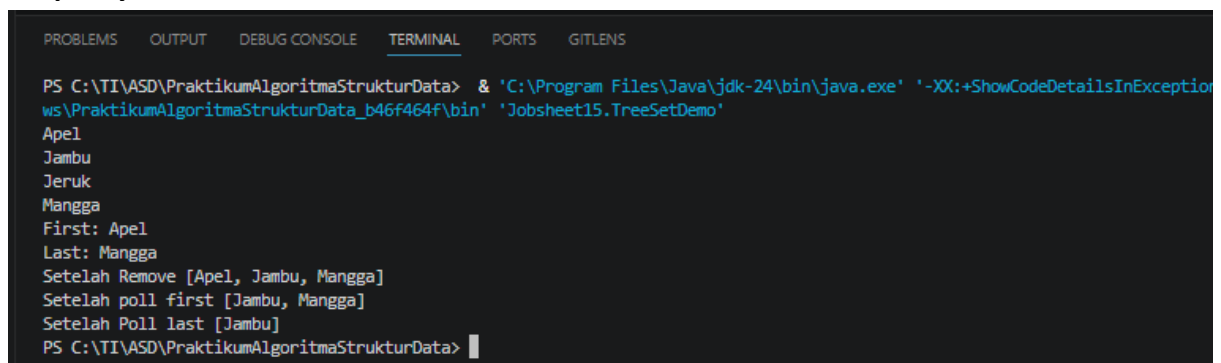
13.5) Praktikum (Implementasi TreeSet).

Code TreeSetDemo.java:



```
1 package Jobsheet15;
2
3 import java.util.TreeSet;
4
5 public class TreeSetDemo {
6     public static void main(String[] args) {
7         TreeSet<String> fruits = new TreeSet<>();
8
9         fruits.add(e: "Mangga");
10        fruits.add(e: "Apel");
11        fruits.add(e: "Jeruk");
12        fruits.add(e: "Jambu");
13
14        for (String temp : fruits){
15            System.out.println(temp);
16        }
17        System.out.println("First: " + fruits.first());
18        System.out.println("Last: " + fruits.last());
19
20        fruits.remove(o: "Jeruk");
21        System.out.println("Setelah Remove " + fruits);
22
23        fruits.pollFirst();
24        System.out.println("Setelah poll first " + fruits);
25
26        fruits.pollLast();
27        System.out.println("Setelah Poll last " + fruits);
28    }
29 }
```

Outputnya:



```
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionThrowing' 'C:\TI\ASD\PraktikumAlgoritmaStrukturData\bin' 'Jobsheet15.TreeSetDemo'
Apel
Jambu
Jeruk
Mangga
First: Apel
Last: Mangga
Setelah Remove [Apel, Jambu, Mangga]
Setelah poll first [Jambu, Mangga]
Setelah Poll last [Jambu]
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>
```

Pertanyaan!

Langkah No.4

Compile dan run program. Mengapa urutan yang ditampilkan berbeda dengan urutan penambahan data ke dalam TreeSet fruits?

Jawaban: Karena dalam TreeSet urutan akan ada dalam bentuk alphabetical mulai dari a (apel), j (jambu, jeruk), m (mangga).

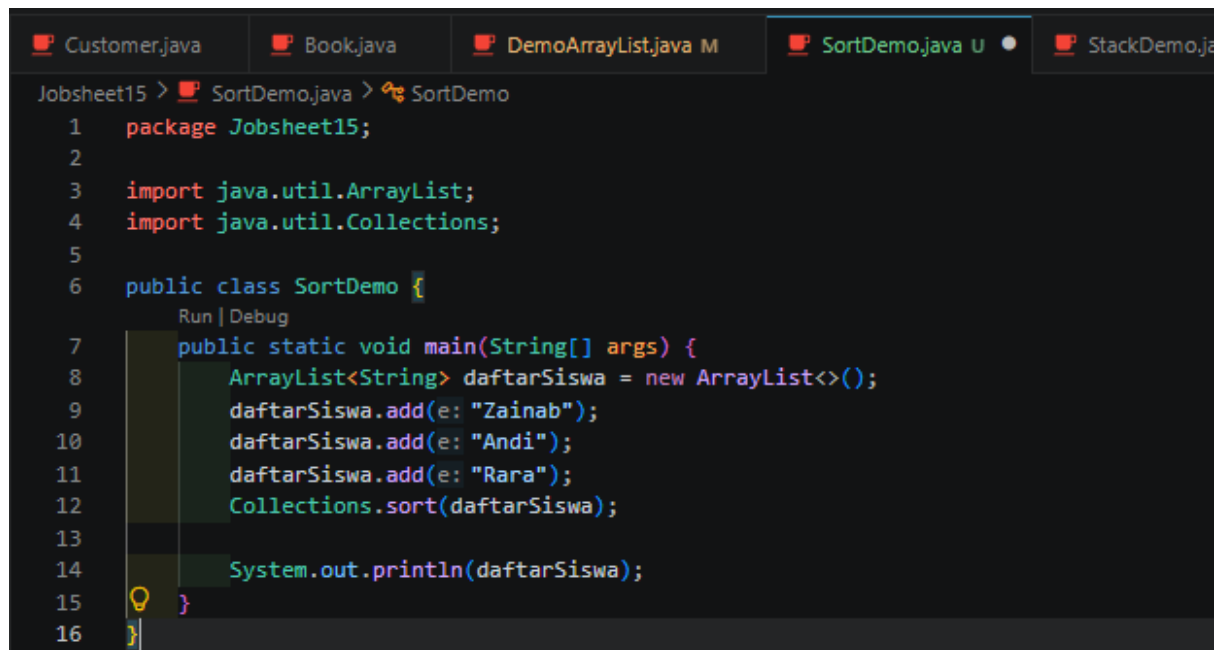
Langkah No.6

Apa yang dilakukan oleh method first(), last(), remove(), pollFirst(), dan pollLast()?

Jawaban: First() akan menampilkan data terdepan, last() akan menampilkan data terakhir, sedangkan pollFirst() akan mengambil data terdepan dan pollLast() akan mengambil data terakhir.

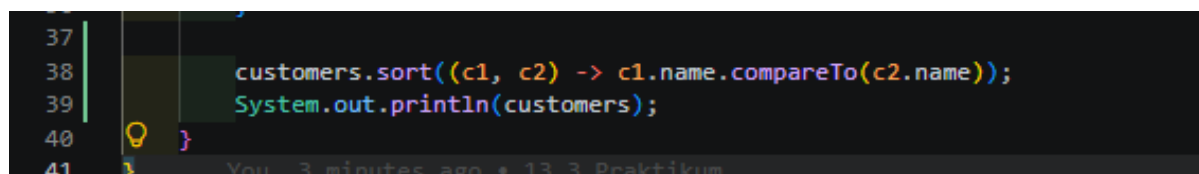
13.6) Praktikum (Sorting).

Code SortDemo.java:



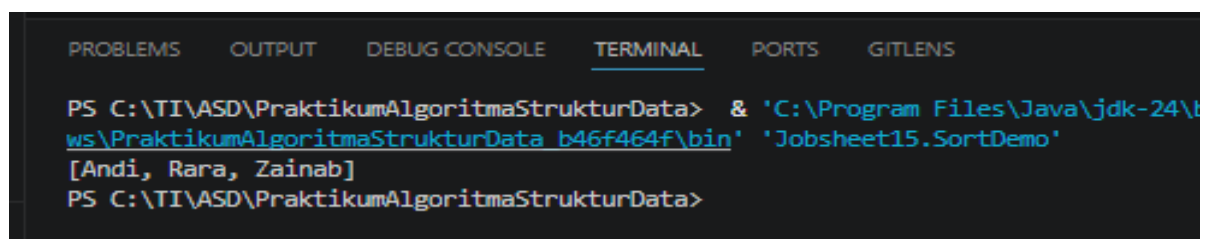
```
Customer.java Book.java DemoArrayList.java M SortDemo.java U StackDemo.java
Jobsheet15 > SortDemo.java > SortDemo
1 package Jobsheet15;
2
3 import java.util.ArrayList;
4 import java.util.Collections;
5
6 public class SortDemo {
7     public static void main(String[] args) {
8         ArrayList<String> daftarSiswa = new ArrayList<>();
9         daftarSiswa.add(e: "Zainab");
10        daftarSiswa.add(e: "Andi");
11        daftarSiswa.add(e: "Rara");
12        Collections.sort(daftarSiswa);
13
14        System.out.println(daftarSiswa);
15    }
16 }
```

Code DemoArrayList.java:



```
37
38 customers.sort((c1, c2) -> c1.name.compareTo(c2.name));
39 System.out.println(customers);
40 }
41 }
```

Outputnya:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData> & 'C:\Program Files\Java\jdk-24\bin\java.exe' -cp 'C:\TI\ASD\PraktikumAlgoritmaStrukturData\bin' 'Jobsheet15.SortDemo'
[Andi, Rara, Zainab]
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>
```